

Options, Black-Scholes, and This Project

Background note — the minimum you need before we start coding.

1. Equities: the foundation

A **stock** (or **equity**) represents a small ownership stake in a company. If a company issues 1 million shares and you own 1,000 of them, you own 0.1% of that company. As a shareholder, you have a claim on the company's future profits (sometimes paid out as dividends). Stocks trade on exchanges, and their price moves constantly based on how investors collectively update their view of the company's future prospects.

For this project, the only things you need to know about a stock are: it has a price today, and that price moves randomly over time. This randomness is not hand-waving — it is the entire mathematical starting point of the Black-Scholes model, formalized as a process called **geometric Brownian motion**, which we'll return to below.

2. What is an option?

An **option** is a contract, not a share of the company itself. It gives its owner the *right, but not the obligation*, to buy or sell a stock at a fixed price, on or before a fixed date. That fixed price is the **strike price** (K), and the fixed date is the **expiry** (T). There are two basic kinds:

- **Call option** — the right to *buy* the stock at the strike price. Worth having if you think the stock is going up: if it rises well above the strike, you can buy it cheap and it's worth more than what you paid.
- **Put option** — the right to *sell* the stock at the strike price. Worth having if you think the stock is going down: if it falls well below the strike, you can sell at the higher strike price instead of the lower market price.

The word "right" (not obligation) is what makes options different from just agreeing to trade later. If the stock moves against you, you let the option expire worthless and lose only what you paid for it (the **premium**). This asymmetry — limited downside, large upside — is what makes an option valuable, and exactly why pricing it correctly is a genuinely hard problem: you're pricing a bet on the future shape of a random process, not just a future price.

One distinction worth knowing: a **European option** can only be exercised on the expiry date; an **American option** can be exercised any time up to expiry. Black-Scholes, in its clean original form, prices European options only — a known limitation, and part of why this project also builds a binomial tree model, which can handle American-style early exercise.

3. What problem is Black-Scholes actually solving?

Suppose a stock trades today at \$100. Someone offers you a call option letting you buy that stock at \$105, three months from now. What is that piece of paper worth, *today*, in dollars? This is genuinely non-obvious — the option's payoff depends on where the stock ends up, a number nobody knows yet.

Black, Scholes, and Merton's insight (1973, later earning a Nobel Prize) was that under specific assumptions about how stock prices move, you can construct a theoretical portfolio — a mix of the stock and a risk-free bond, continuously rebalanced — that exactly replicates the option's payoff no matter what the stock does. If so, the option must cost the same as building that portfolio today; otherwise a trader could earn a riskless profit from the mismatch. This **no-arbitrage** argument is what pins the price to a single, exact number.

The key assumptions are worth knowing explicitly, since a large part of this project (the implied-volatility and delta-hedging stages) is about seeing exactly where they break down in real markets: the stock follows a continuous random path with *constant* volatility (no jumps); markets have no transaction costs and allow trading in any quantity, any time; there's a known, constant risk-free rate; and the option is European-style. None of

these hold exactly in real markets — and that gap is precisely what a quant researcher gets paid to understand.

3.1 The governing equation

The replicating-portfolio argument, made rigorous, leads to a partial differential equation (PDE) that any option's price $V(S, t)$ must satisfy:

$$\frac{\partial V}{\partial t} + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + r S \frac{\partial V}{\partial S} - r V = 0$$

You won't need to solve this PDE by hand — its role here is conceptual: this is the actual "Black-Scholes equation," and the closed-form formula below is its solution for a standard European call or put.

3.2 The closed-form pricing formula

For a European call option:

$$C = S_0 N(d_1) - K e^{-rT} N(d_2)$$

For a European put option:

$$P = K e^{-rT} N(-d_2) - S_0 N(-d_1)$$

where:

$$d_1 = \frac{\ln(S_0 / K) + (r + \sigma^2 / 2)T}{\sigma \sqrt{T}}$$

$$d_2 = d_1 - \sigma \sqrt{T}$$

and $N(\cdot)$ is the standard normal cumulative distribution function — $N(x)$ is the probability a standard normal random variable is less than x .

The five inputs, in plain terms:

- S_0 — the stock's current price
- K — the strike price
- T — time remaining until expiry (in years)
- r — the risk-free interest rate
- σ (sigma) — the stock's volatility: an annualized standard deviation of returns

Four of these — S_0 , K , T , r — are directly observable in the market right now. Only σ is not; it must be estimated or assumed. This single fact seeds the entire implied-volatility stage of this project: since everything else in the formula is known, you can run it "backwards" — take a real market option price and solve for whatever σ would make the formula match it. That backed-out number is **implied volatility**: what level of future risk the market is pricing in, as opposed to what's been observed historically.

3.3 A useful sanity check: put-call parity

A model-independent relationship worth knowing — it follows from no-arbitrage alone, without the full formula:

$$C - P = S_0 - Ke^{-rT}$$

Once you implement your pricer, checking that computed call and put prices satisfy this exactly is a fast, strong correctness test.

4. The Greeks, briefly

The Greeks are the option price's sensitivities to each input — the partial derivatives of the formula above. They matter practically because they tell you exactly how a position's value shifts as market conditions move, which is the basis of hedging (Section 6). This project implements: **Delta** (sensitivity to the stock price — also the exact hedge ratio used in Section 6), **Gamma** (sensitivity of delta itself, i.e. how often a hedge needs rebalancing), **Vega** (sensitivity to volatility — connects directly to the implied-volatility work), **Theta** (value lost purely from time passing), and **Rho** (sensitivity to interest rates, usually the least important in practice).

5. Why the model is wrong, on purpose, and why that's the point

Black-Scholes rests on assumptions that don't hold exactly in real markets: volatility isn't actually constant, it changes over time and spikes during market stress; real stock prices occasionally jump rather than moving continuously; trading isn't free or infinitely divisible. This project isn't an attempt to "prove Black-Scholes right." The value of building it yourself, from the equations up, is that it puts you in a position to precisely locate *where and how* the model's clean assumptions diverge from reality — and explaining that divergence clearly, with evidence, is a far stronger signal of understanding than reciting the formula from memory. That's the through-line connecting every stage below.

6. What we're actually building, stage by stage

- **Stage 1 — Implement Black-Scholes and the Greeks from the equations above**, not a library's built-in pricer, then check the results against put-call parity and known edge cases.
- **Stage 2 — Cross-validate with Monte Carlo and a binomial tree.** Monte Carlo simulates thousands of random stock paths directly and should converge to the same price as the closed-form formula — a strong independent check. The binomial tree additionally handles American-style early exercise, which Black-Scholes cannot.
- **Stage 3 — Pull real option prices and back out implied volatility**, then plot it across strikes and expiries to see the "volatility smile" — direct evidence of where the constant-volatility assumption breaks down.
- **Stage 4 — Simulate delta-hedging on real historical prices** and measure the resulting profit and loss. In a perfect Black-Scholes world this hedge would be exact; in practice, discrete rebalancing and volatility mismatches produce real P&L; — a concrete, measurable demonstration of the model-vs-reality gap from Section 5.

This note is background only — no code yet. Once this is comfortable, Stage 1 starts with implementing the formulas in Section 3.2 directly in Python.